Pandelis Zembashis

CTO | AI Runtime Systems | Inference, Performance & Compiler-Adjacent Engineering

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EXECUTIVE SUMMARY

CTO and hands-on systems engineer building AI runtimes, execution systems, and performance-sensitive developer tooling. At Cosine, led the Go runtime and harness behind coding agents: graph-like tool orchestration, deterministic replay, repository-scale search, WebAssembly portability, Kubernetes/gVisor execution, and structured evaluation signals for model improvement. Relevant to NVIDIA LPX inference/compiler work through runtime design, workload mapping, profiling-driven optimization, concurrency, graph execution, and feedback between model behavior, software systems, and compute constraints.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Applied AI lab building coding-agent runtimes, model feedback loops, and secure execution infrastructure.

- Led product and engineering for Cosine's AI coding platform, spanning runtime architecture, model/tool execution, performance, observability, secure compute, and research-to-product feedback loops.
- Built Cosine's Go harness for agent execution: event scheduling, streaming tool calls, checkpointing, deterministic replay, memory writes, MCP/LSP integrations, subagents, file locks, and structured rollout traces.
- Reworked filesystem and repository-ingestion hot paths around native Go concurrency, bundled ripgrep-class search, streaming file reads, low-overhead tool results, and predictable performance on large codebases.
- Shipped one runtime across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, keeping execution semantics stable across local, embedded, and cloud-backed environments.
- Built sandboxed execution from 2024 onward: Kubernetes-scheduled agents, gVisor/runsc isolation, ephemeral sessions, network-policy boundaries, and secure shell/tool access for model-controlled workloads.
- Built evaluation and profiling signals around the runtime: replayable traces, tests, LSP diagnostics, checkpoints, reviews, and parallel sandboxed runs to understand model/tool behavior under real workloads.
- Helped position Cosine as an AI lab and product company with benchmark-leading coding models, UK sovereign AI backing, and 500,000 AIRR GPU hours on Isambard-AI.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established engineering cadence, incident response habits, and onboarding rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving product iteration across web, operations, and marketplace workflows.
- Established development cadences and product intake processes that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

SELECTED TECHNICAL WORK

Why we built our own harness: Go runtime design, tool execution, replay, orchestration, gVisor/Kubernetes execution, and model feedback loops for coding agents.
Cosine CLI runtime: Single-binary distribution, native search, filesystem performance, and low-friction model/tool runtime deployment.
Tech.eu on Cosine: Coverage of Cosine's benchmark-leading coding models, sovereign AI work, and full-stack model/product strategy.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Runtime & Performance Systems: Go, Python, concurrency, streaming I/O, repository indexing, native search, profiling signals, replay, checkpointing
Model Execution Infrastructure: Agent runtimes, graph-like orchestration, tool scheduling, evaluation traces, model feedback loops, LSP/test signals, WebAssembly
Systems & Compute Platforms: Kubernetes, gVisor/runsc, containers, Linux systems, AWS, secure execution, resource scheduling, observability, CI/CD
Compiler-Adjacent Foundations: Data structures, algorithms, graph execution, IR-style event models, workload mapping, benchmarking, debugging, technical communication